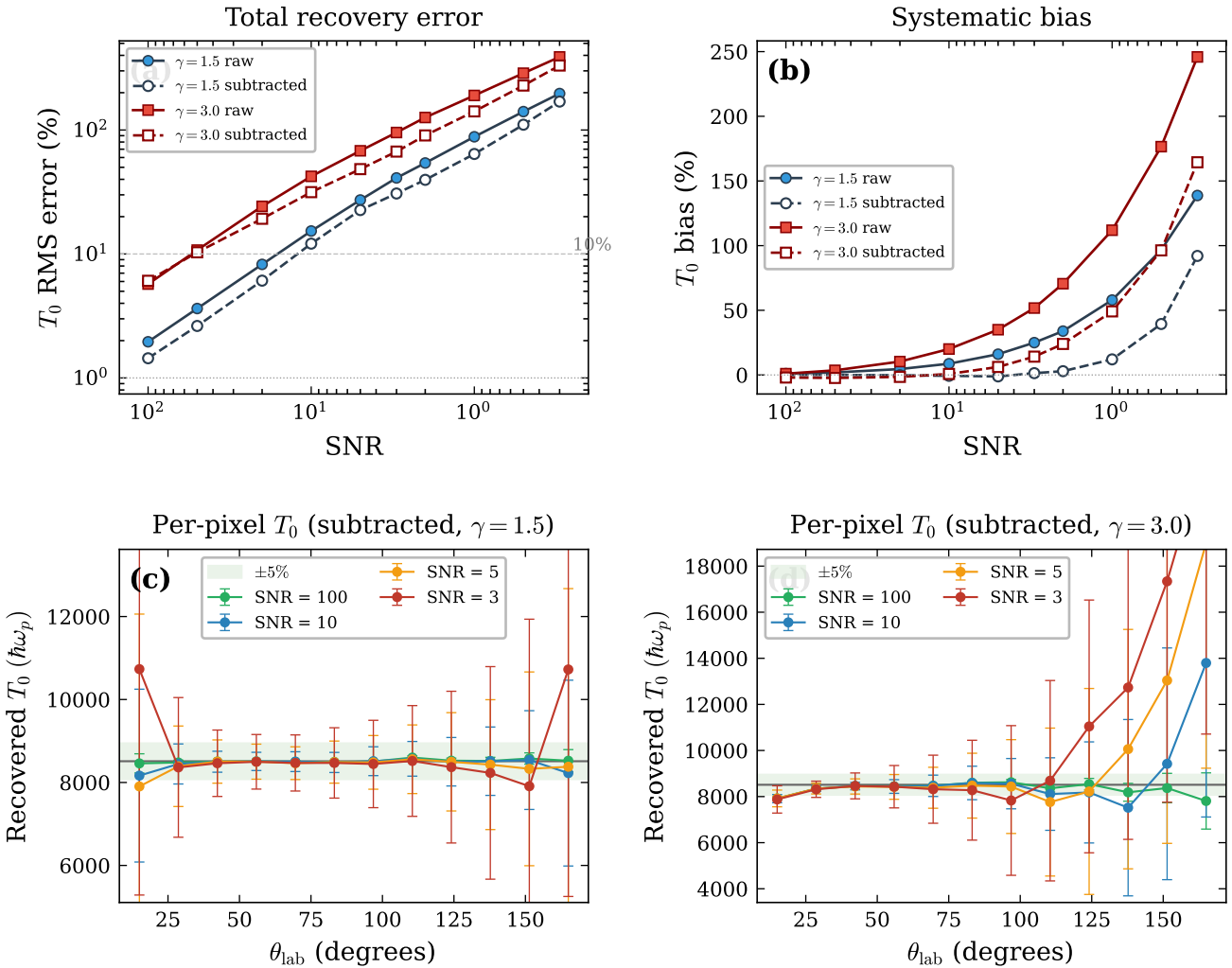

Noise susceptibility: additive detector noise on ensemble-averaged PSD
$T_0 = 8512\,\hbar\omega_p$, 1000 noise trials per SNR, $N_{\rm ens} = 1000$ clean realizations
Total recovery error
(a)
$\gamma = 1.5$ raw
$\gamma = 1.5$ subtracted
$\gamma = 3.0$ raw
$\gamma = 3.0$ subtracted
10%
T_0 RMS error (%)
SNR
Systematic bias
(b)
T_0 bias (%)
SNR
Per-pixel T_0 (subtracted, $\gamma = 1.5$)
(c)
±5%
SNR = 100
SNR = 10
SNR = 5
SNR = 3
Recovered T_0 ($\hbar\omega_p$)
$\theta_{\rm lab}$ (degrees)
Per-pixel T_0 (subtracted, $\gamma = 3.0$)
(d)
±5%
SNR = 100
SNR = 10
SNR = 5
SNR = 3
Recovered T_0 ($\hbar\omega_p$)
$\theta_{\rm lab}$ (degrees)